

Homework 2: Submission 3

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Question 1

The distribution of plan counts by county is visualized in Figure 1, where the median of the plan count is shown to increase over time:

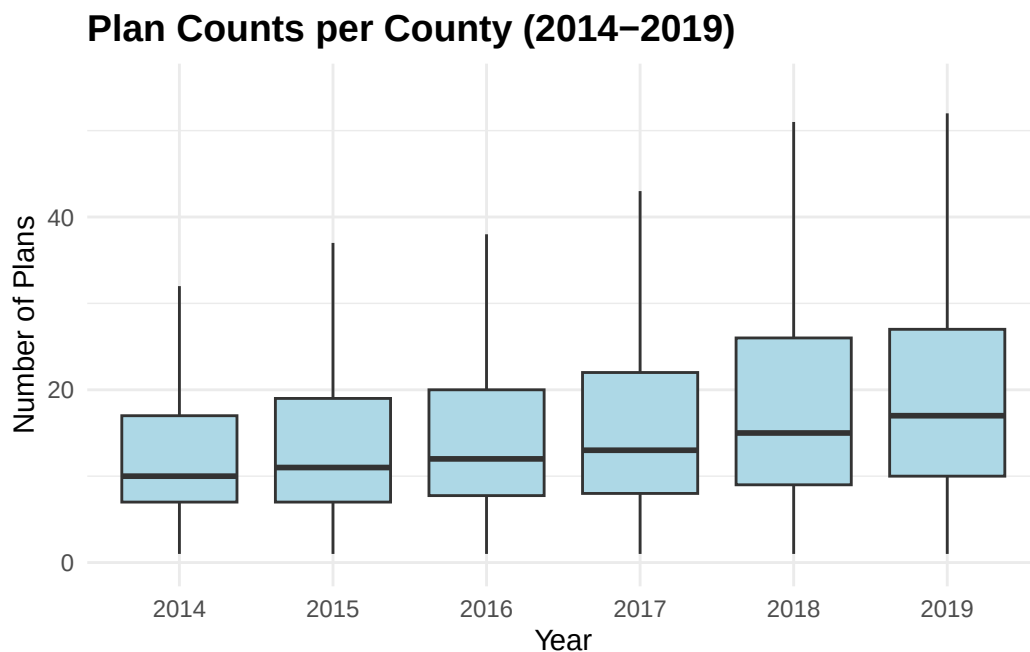


Figure 1: Distribution of Plan Counts per County (2014-2019)

An increase in the median of insurers from 10 insurers per county in 2014 to 17 insurers in 2019 would suggest that the market competition is **sufficient** for beneficiaries to select from a wide variety of plans with various benefits, premiums, and provider networks. More insurers and lower concentration also leads to lower cost-sharing prices and smaller markups that generally favor enrollees.

However, based on the distribution, arguments for beneficiaries in a given county having too many or too few plans can also be justified. For instance, some counties across all six years have access to over 50 plans. Especially for elderly and disabled cohorts that utilize Medicare Advantage (MA), it is often impossible to select from a large catalog of plans that best fits their financial and health circumstances. Beneficiaries are, thus, highly vulnerable to defaulting towards a low-value plan

that is not well-informed, despite the large number of plans ostensibly providing a greater freedom of choice.

Conversely, while this has improved over the years, some counties have less than 10 plans, represented by the lower quartiles of the plots. This may reflect a geographical disparity in the availability of insurers, as beneficiaries residing in rural areas tend to have less access to both health insurance plans and providers compared to urban dwellers. Nevertheless, more data analysis is necessary to support this conclusion.

Question 2

The distribution of MA plan bids is shown in Figure 2:

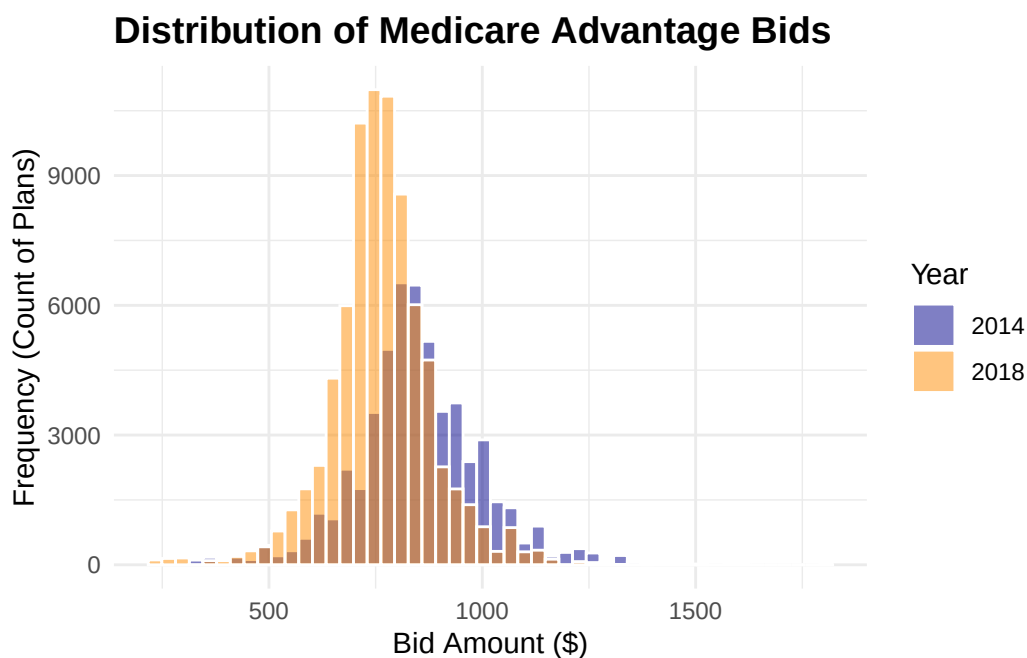


Figure 2: Distribution of Medicare Advantage Bids (2014 vs 2018)

When comparing the distributions between 2014 and 2018, it is remarkably clear that there are more plans in total in 2018 compared to 2014. This supports the previous assertion that the plan count has increased over time.

Another shift to note is that the peak of the distribution shifted slightly to the left in 2018 compared to 2014. This may signal that, as more plans entered the market, insurers felt the pressure of market competition to lower their bids, increasing the difference from the benchmark to offer higher rebates to beneficiaries.

Lastly, the distribution of bids in 2018 is narrower compared to 2014, indicating that the health insurance market may be shifting closer to efficiency as insurers standardize their bids and, subsequently, the benefits they provide to enrollees to stay competitive.

Question 3

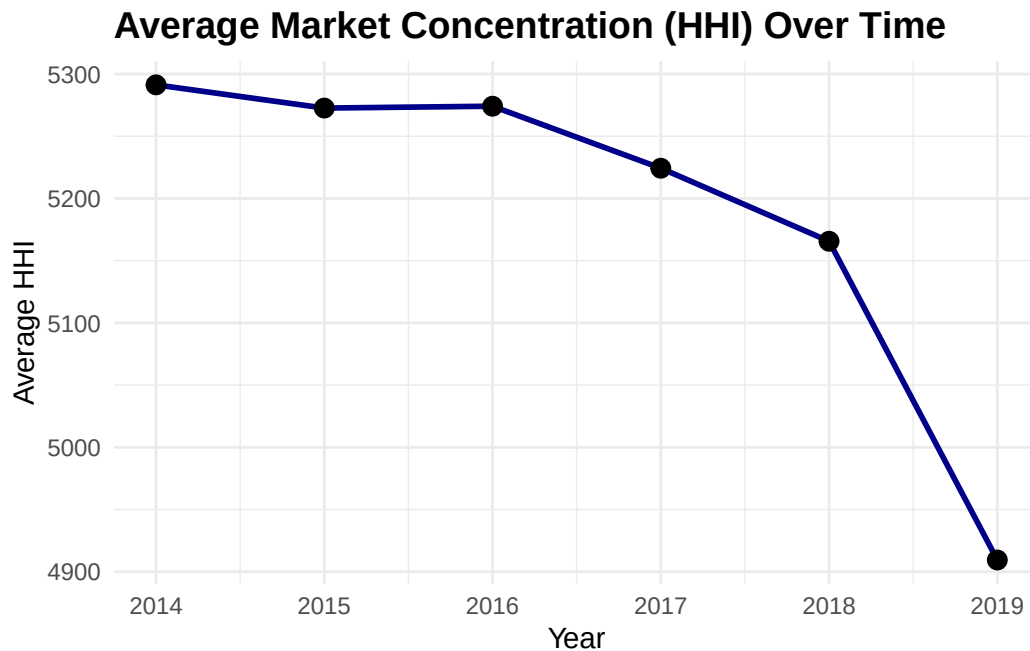


Figure 3: Average Market Concentration (HHI) Over Time (2014-2019)

As mentioned in class, the Herfindahl-Hirschman Index (HHI) measures market concentration, with 0 representing perfect competition and 10,000 representing a complete monopoly.

Based on Figure 3, the average HHI across all counties remained stable at about 5,270 from 2014 to 2016, indicating that the health insurance market was highly concentrated and structured oligopolistically (a few large insurers dominate the industry). The average HHI began to decrease in 2016, with the largest decrease taking place between 2018 (5,166) and 2019 (4,910). However, it is important to note that the national health insurance market is still highly concentrated across all years, with the average HHI consistently staying above the Federal Trade Commission's (FTC) threshold for "highly concentrated" markets of 2,500.

Question 4

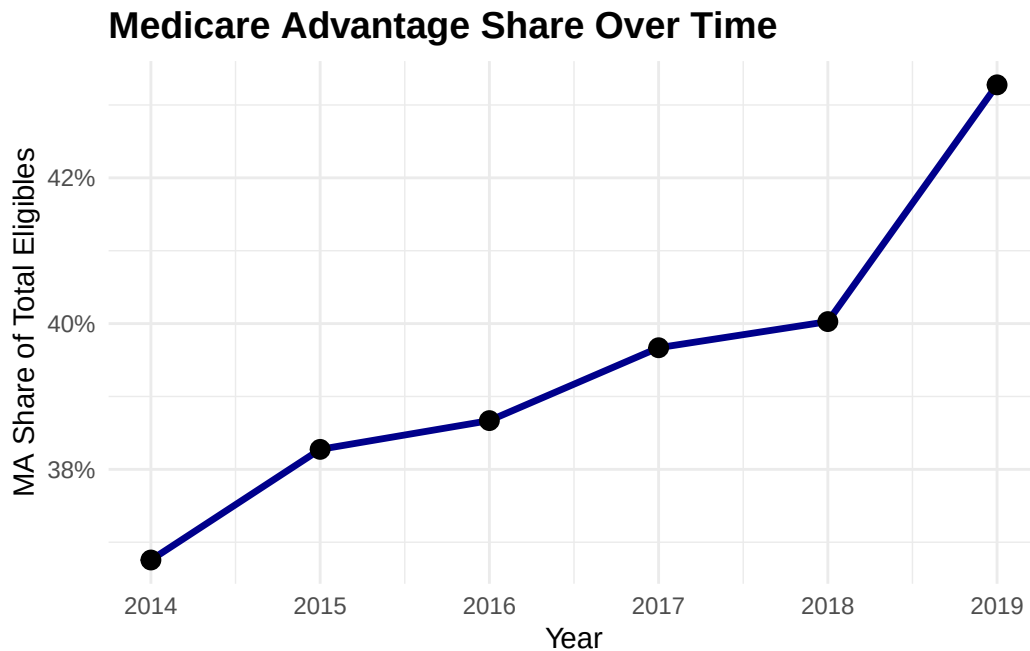


Figure 4: Medicare Advantage Share Over Time (2014-2019)

According to Figure 4, the average share of MA enrollees compared to all Americans that are eligible for Medicare increased from 36.8% in 2014 to 43.3% in 2019, indicating that MA has become more popular for Medicare eligibles over the years.

Question 5

Table 1: Mean Medicare Advantage Bids, FFS Costs, and Medicare Eligibles by HHI (2018)

Variable	Competitive (Low HHI)	Uncompetitive (High HHI)	Overall
Plan Bid	\$781.99	\$786.90	\$784.48
FFS Cost	\$9,562.20	\$9,128.32	\$9,341.97
Medicare Eligibles	27,027	11,723	19,259

According to Table 1, the mean bids are slightly higher in uncompetitive markets (high HHI) than competitive markets (low HHI), which may indicate lower competitive pressure and more freedom to exercise market power to charge higher prices for enrollees. Going back to the argument of geographical disparities, rural areas tend to have fewer beneficiaries, which increases the risk of adverse selection from these relatively small risk pools.

Interestingly though, FFS costs are lower in concentrated markets compared to competitive markets. This suggests that, while insurers in competitive markets are submitting lower bids towards

the actual cost of care to offer extra, more enticing benefits for beneficiaries, insurers in uncompetitive markets, with no market rivals to undercut them, are submitting higher bids to pocket a greater share of Medicare benchmarks as profit. Consequently, costs are passed onto the beneficiaries, which may help to explain, in part, the health disparities observed in sparsely populated areas compared to densely populated ones.

Question 6

Table 2: Average Bids by FFS Cost Quartile and Market Competition

FFS Cost Quartile	Competitive (Low HHI)	Uncompetitive (High HHI)
Quartile 1 (Low Cost)	\$792.61	\$798.75
Quartile 2	\$779.88	\$784.29
Quartile 3	\$776.66	\$774.55
Quartile 4 (High Cost)	\$781.10	\$787.74

According to Table 2, bids in quartile 1, which includes counties with the lowest FFS costs, are highest out of all other quartiles. This challenges initial expectations that insurers in counties with the lowest FFS costs will submit lower bids. However, it also corroborates the assumption that insurers submit higher bids as a cushion between high Benchmarks and low healthcare costs to capture more profit.

It would be counterintuitive to submit higher bids in quartile 4, which includes counties with the highest FFS costs, than quartile 1 since there would be less rebates to return to the beneficiary, especially in competitive markets where there is more financial incentive to keep plans attractive. To have a competitive rebate margin, submitting efficient bids is necessary, though this is not reflected in uncompetitive markets, where insurers have more market power to submit higher bids regardless of FFS costs.

Question 7

Table 3: ATE Estimates

Estimator	ATE
NN Matching (Inv. Var.)	\$3.77 (2.6)
NN Matching (Mahalanobis)	\$3.77 (2.6)
IPW Weighting	\$3.77 (2.59)
OLS with Interactions	\$3.77 (2.6)

Nevertheless, the average treatment effects (ATE) shown in Table 3 highlight that price gouging by insurers in concentrated markets may not be intentionally for profit. While point estimates across all estimators suggest that uncompetitive markets have bids that are \$3.77 higher, when compared with their respective standard errors, the results are statistically indistinguishable from zero. This indicates, then, that structural challenges inherent in concentrated markets, such as higher administrative costs per beneficiary, may play a bigger role in the difference in observed bids than attempts to capture higher revenues.

Question 8

The ATEs (\$3.77) are identical across all estimators in question 7.

Question 9

Table 4: OLS Comparisons

Model	ATE
OLS Baseline	\$3.77 (2.6)
OLS Alternate	\$2.60 (2.62)

When controlling for total Medicare beneficiaries in Table 4, the lack of statistical significance demonstrated in Table 3 becomes even more apparent. When the ATE is re-estimated using continuous FFS costs, the causal effect of HHI on bids becomes even smaller compared to the ATEs calculated using discrete FFS cost quartiles.

Ultimately, this suggests that there is no causal relationship between market competition and bids.

Question 10

Overall, I thought that the process of visualizing data was very fun. Switching from Excel, where I worked with pre-designed plots, to R where I have much more flexibility with designing the charts was exciting. Since I am taking Econometrics alongside this course, much of the conceptual material we learn in class is still very new to me. So, I spend much more time outside of class than my fellow classmates just learning the concepts. However, this assignment taught me that simple plots and histograms don't really tell the whole story and that estimators tell the truth much clearer.

I think a lot of my classmates will agree with me on this when I say that the number of times I had to go back and fix my code was the most aggravating part of the assignment. It should be illegal

for a single typo to break an entire block of code. Why R Studio doesn't have a built-in Grammarly (but for code) is something I will never understand. I feel sorry for computer science majors...